TWIN NODE: A LOW-POWER WIRELESS WATER LEVEL MONITORING SYSTEM USING ESP-NOW FOR OVERHEAD TANKS

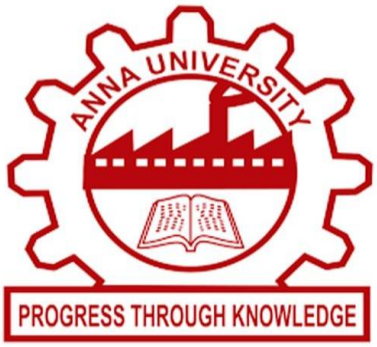
A PROJECT REPORT

[EC23C21] - MICROPROCESSORS AND MICROCONTROLLERS

Submitted by

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OCTOBER 2026

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1. Objective of the Project

The objective of the TWIN NODE project is to design, build and test a low-cost, low-power wireless system that monitors an overhead tank's water level and informs the user indoors without any router, cloud or internet. It has an outdoor sensor node (TWIN NODE OUTDOOR) and an indoor display node (TWIN NODE INDOOR). The specific objectives are:

- To measure the water level contactlessly with a waterproof ultrasonic sensor.
- To transfer the measurement between two ESP8266 (ESP-12F) nodes using ESP-NOW.
- To indicate the level with four LEDs and sound a buzzer when the tank is full or empty.
- To power the outdoor node from a solar panel and 18650 Li-ion cell, and the indoor node from a USB-charged Li-ion cell.
- To design both nodes as KiCad PCBs and to fabricate and test the outdoor node

2. Abstract

Manual checking of overhead tanks causes overflow, wasted electricity and dry running of pumps. TWIN NODE uses two ESP-12F-based nodes that communicate directly over ESP-NOW. The outdoor node measures the distance to the water surface with an ultrasonic sensor, converts it into a percentage and level, and transmits it every 30 seconds. It runs on a solar panel, TP4056 charger, 18650 cell with DW01A/FS8205A protection and an RT9080 3.3 V regulator, and sleeps between readings.

The indoor node accepts packets only from the known transmitter, lights one of four LEDs for the level and drives a buzzer for the full and empty conditions. The outdoor board has been fabricated and powered on successfully, both boards pass the KiCad DRC with zero violations, and a relay-based motor controller is identified as future work.

3. Methodology

3.1 System Overview

The system follows a transmitter–receiver architecture: NODE-1 senses and transmits, and NODE-2 receives and alerts. ESP-NOW exchanges short packets directly between MAC addresses at 2.4 GHz, so no access point is needed.

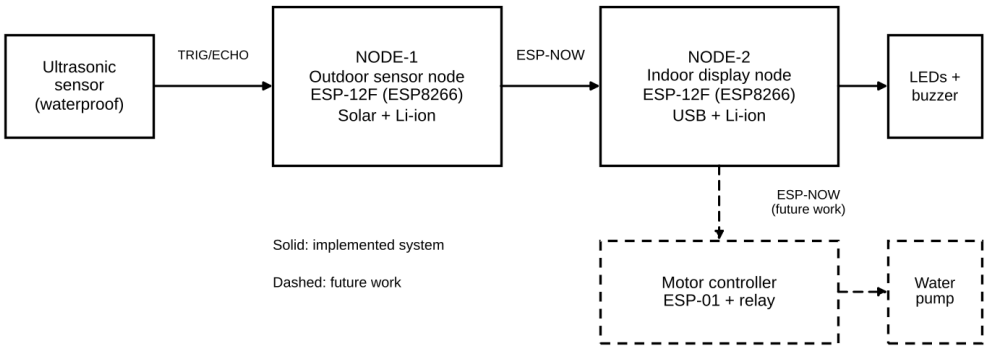


Figure 3.1 Block diagram of the TWIN NODE system (the motor controller is future work)

3.2 Design Methodology

1. **Requirements:** no internet, solar and battery operation, simple indoor alerts, manufacturable boards.

2. **Architecture and components:** ESP-12F with ESP-NOW, ultrasonic sensing, TP4056 with DW01A/FS8205A, RT9080 LDO.
3. **Schematic and PCB design:** KiCad schematics checked with ERC and layouts with DRC; the board evolved from V1.0 to V2.0.
4. **Firmware:** Arduino-framework code with callbacks and non-blocking timing for each node.
5. **Fabrication and testing:** the V2.0 outdoor board was fabricated, powered on and its boot and reset behaviour validated.

Tools: KiCad EDA 9.0.x, Arduino framework for ESP8266, and Git/GitHub (MAATHES-THILAK-K/TWIN-NODE).

4. System Architecture

4.1 Outdoor Node Architecture (TWIN NODE OUTDOOR)

The outdoor node is the fabricated and tested V2.0 revision, a two-layer board of about 45 × 45 mm with power, controller, sensing and interface sections (Figure 4.1).

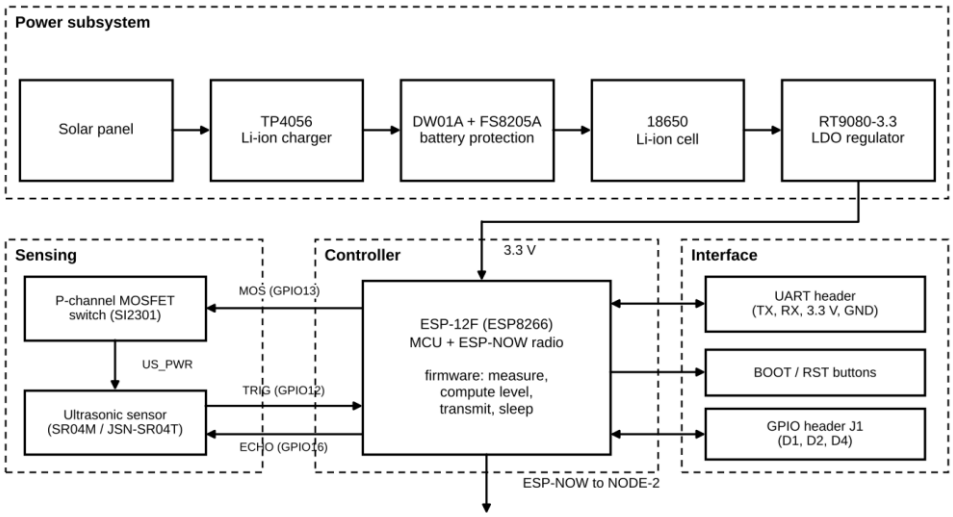


Figure 4.1 Hardware architecture of the outdoor sensor node

Table 4.1 Outdoor node architecture summary

Layer	Implementation
Sensing	Waterproof ultrasonic sensor (SR04M / JSN-SR04T). TRIG on GPIO12, ECHO on GPIO16. Sensor supply (US_PWR) is switched by an SI2301 P-channel MOSFET controlled from GPIO13 (active LOW).
Control	ESP-12F (ESP8266) with boot-strap resistors, BOOT and RST push-buttons with 100 nF capacitors, and 100 µF + 10 µF decoupling on the 3.3 V rail.
Communication	ESP-NOW on channel 1, unicast to the indoor node MAC address, with a send-status callback.
Power	Solar panel, TP4056 charger (blue charge LED), DW01A + FS8205A protection, 18650 Li-ion cell, RT9080-3.3 LDO.
Interface	JST connectors for battery, solar panel, sensor and UART (3.3 V, TX, RX, GND); GPIO header with D1, D2 and D4 (the firmware uses GPIO14 for an alarm buzzer driver).

Layer	Implementation
Software layers	Drivers (GPIO, pulse timing) → measurement (distance, percentage, level) → communication (ESP-NOW and callback) → power manager (sensor MOSFET and timed light sleep) → alarm (non-blocking beep pattern).

4.2 Indoor Node Architecture (TWIN NODE-INDOOR)

The indoor node is a receiver and alert unit built on the same ESP-12F, charger, protection and regulator blocks, with the sensing section replaced by an indication and alert section (Figure 4.2).

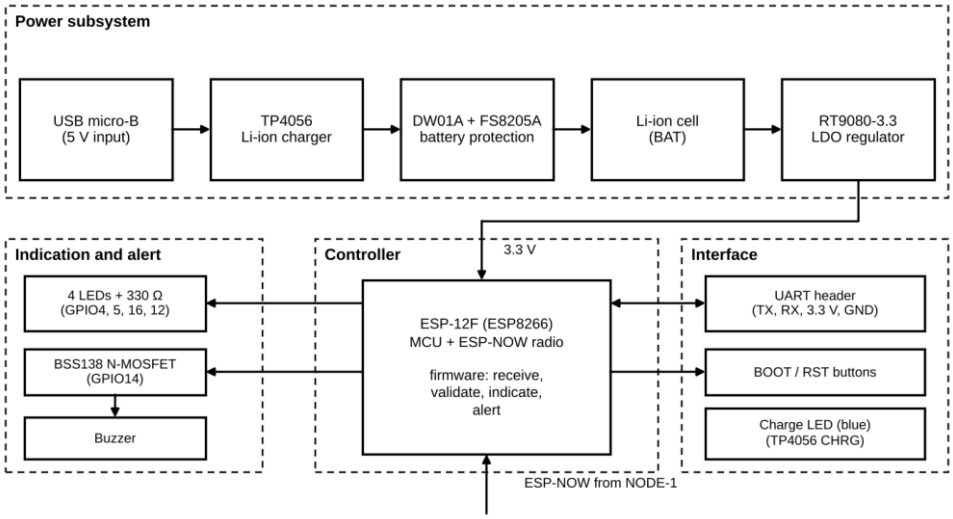


Figure 4.2 Hardware architecture of the indoor display node

Table 4.2 Indoor node architecture summary

Layer	Implementation
Communication	ESP-NOW receiver on channel 1 in the slave role; accepts packets only from the known transmitter MAC address and of the correct size.
Control	ESP-12F (ESP8266) with boot-strap resistors and RST and BOOT buttons (10 kΩ and 100 nF each).
Indication	Four LEDs (L1 to L4) with 330 Ω resistors on GPIO4, GPIO5, GPIO16 and GPIO12 show the water level.
Alert	BSS138 N-channel MOSFET on GPIO14 switches an external buzzer connected between 3.3 V and NEG_BUZZ.
Power	Micro-USB 5 V input, TP4056 charger (blue charge LED), DW01A + FS8205A protection, Li-ion cell, RT9080-3.3 LDO, 100 µF bulk capacitor.
Software layers	ESP-NOW receive callback (validation) → status variables (valid, full, empty, percentage) → LED driver → non-blocking buzzer scheduler in the main loop.

5. Circuit Diagram

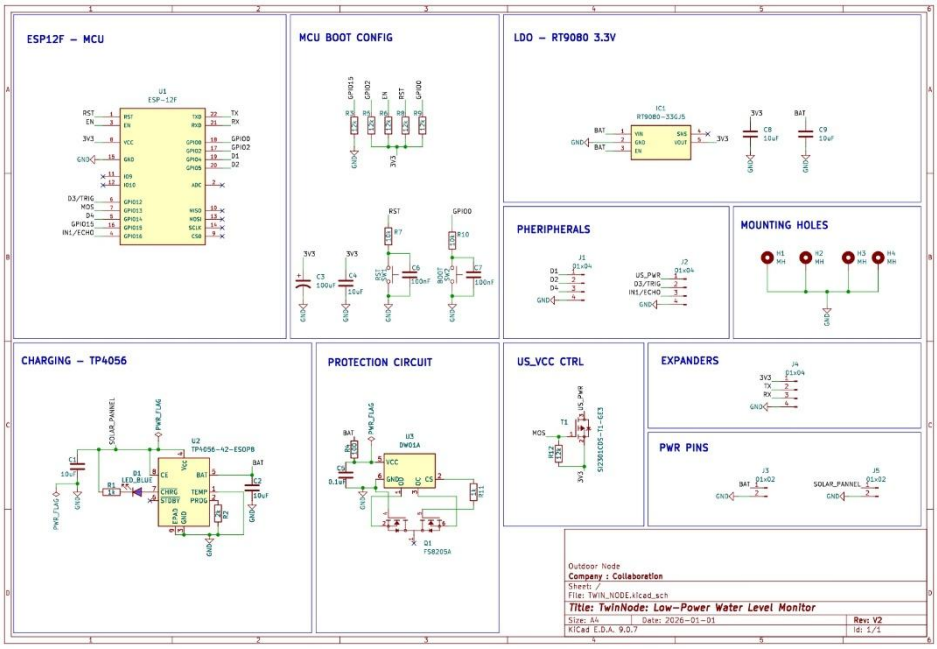


figure 5.1 Circuit diagram of the outdoor sensor node (TWIN NODE OUTDOOR)

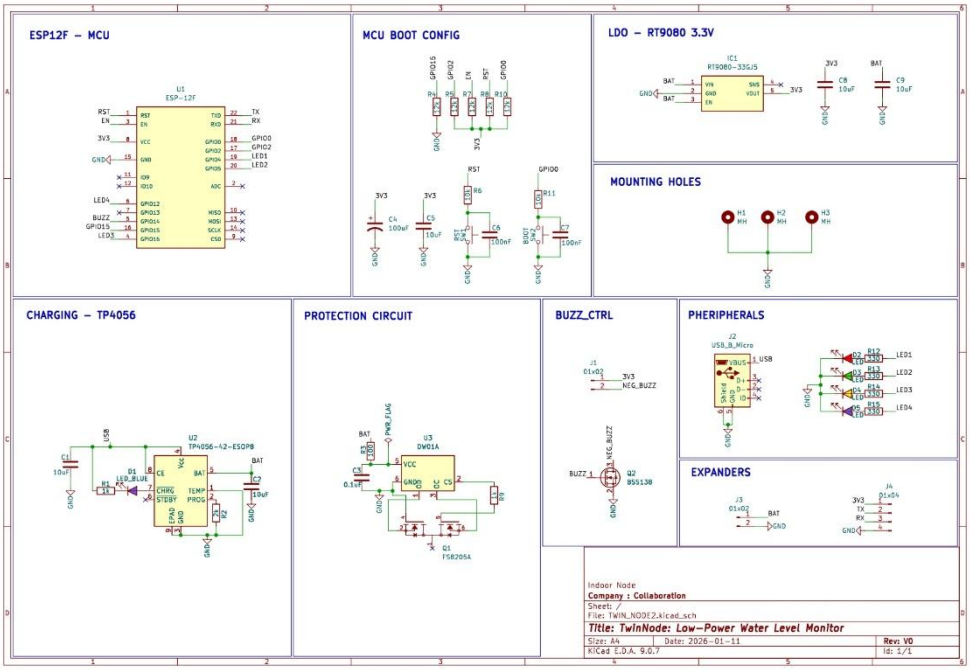


Figure 5.2 Circuit diagram of the indoor display node (TWIN NODE INDOOR)

6. Circuit Explanation

6.1 Outdoor Node

Microcontroller and boot configuration. GPIO12 is the ultrasonic trigger, GPIO16 the echo input and GPIO13 the sensor-supply control. At reset, 12 kΩ resistors hold GPIO15 low and GPIO2, EN, RST and GPIO0 high for normal flash boot. The BOOT and RST buttons pull GPIO0 and reset low, each with a 100 nF capacitor.

Power chain. The solar panel feeds the TP4056, whose 2 kΩ PROG resistor sets the charge current and whose CHRG pin drives a blue LED. The charger output connects to the 18650 cell through the DW01A/FS8205A protection, which cuts the cell off on over-charge, over-discharge or over-current. The RT9080-3.3 LDO, chosen for its very low quiescent current, powers the 3.3 V rail, and a 100 µF capacitor supplies the radio current bursts.

Sensor power switch. The SI2301 P-channel MOSFET connects 3.3 V to US_PWR. A 12 kΩ resistor keeps the sensor OFF by default and GPIO13 pulls the gate low to turn it ON, removing standby current during sleep.

Connectors. J1 (D1, D2, D4, GND), J2 (sensor), J3 (battery), J5 (solar panel) and J4 (UART). Four mounting holes are tied to ground.

6.2 Indoor Node

Controller and indicators. The indoor ESP-12F uses the same boot-strap network as the outdoor node. Four LEDs (each with a 330 Ω resistor) are driven directly from GPIO4, GPIO5, GPIO16 and GPIO12.

Buzzer driver. GPIO14 drives a BSS138 N-channel MOSFET that switches the low side of an external buzzer, keeping the buzzer current off the GPIO pin.

Power. A micro-USB connector supplies 5 V to the TP4056 charger. The cell is protected by a DW01A and FS8205A, and an RT9080-3.3 LDO generates the 3.3 V rail. A battery connector and a UART header complete the board.

7. Working Methodology

7.1 Distance and Level Measurement

The outdoor firmware sends a 10 µs pulse on TRIG and measures the echo width t (µs) with a 30 ms timeout. The distance and water percentage are:

$$\textit{Distance (cm)} = (t \times 0.0343) / 2$$

$$\textit{Water \%} = ((25 - \textit{Distance}) / (25 - 5)) \times 100, \textit{ limited to 0–100}$$

A smaller distance means a fuller tank. If no echo is received, the reading is flagged invalid (−1) and never treated as empty. Table 7.1 gives the level classification.

Table 7.1 Level classification in the outdoor node firmware

Distance to water	Level	Status	Flags sent
≤ 5 cm	4	FULL	tankFull = true
> 5 and ≤ 12 cm	3	HIGH	–
> 12 and ≤ 20 cm	2	MEDIUM	–
> 20 and ≤ 25 cm	1	LOW	–
> 25 cm	0	EMPTY	tankEmpty = true

Distance to water	Level	Status	Flags sent
No echo	0	SENSOR ERROR	distance = −1, no alarm

7.2 Firmware Operation

Figure 7.1 shows the firmware flow for both nodes. The ESP-NOW packet holds distanceCm, waterPercent, level, tankFull and tankEmpty; the receiver rejects a packet of a different size.

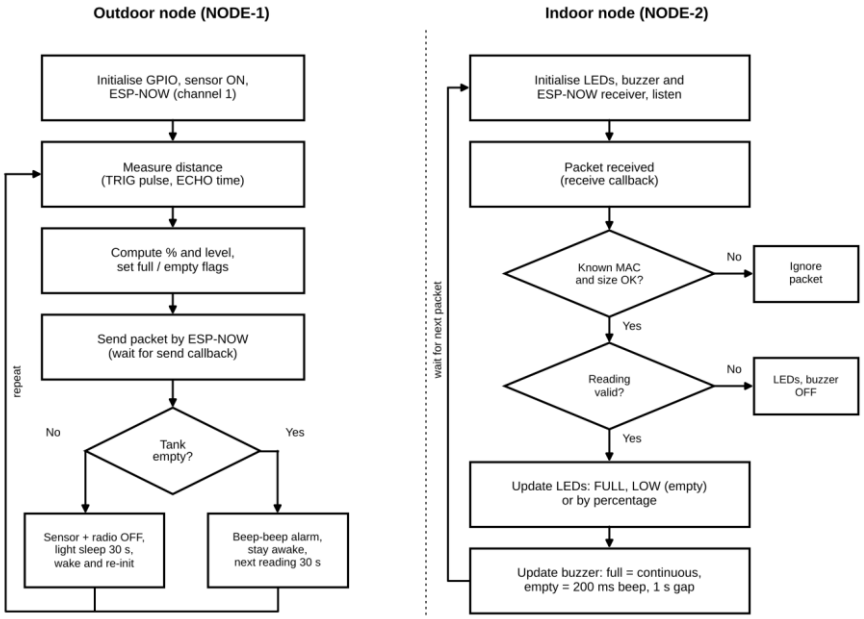


Figure 7.1 Firmware flowcharts of the outdoor and indoor nodes

Outdoor node. It measures, computes the percentage and level, and sends the packet every 30 seconds. If the tank is empty, it stays awake and runs a non-blocking beep pattern. Otherwise it switches off the sensor supply and radio and enters a 30-second timed light sleep, then restores the sensor, waits 500 ms and re-initialises ESP-NOW. Light sleep is used because GPIO16 is needed as the echo input.

Indoor node. The receive callback checks the sender MAC address and packet size and accepts only non-negative readings. The LEDs show FULL when the full flag is set, LOW when the empty flag is set, and otherwise LOW (up to 25 %), MEDIUM (up to 50 %), HIGH (up to 80 %) or FULL (above 80 %). The buzzer is continuous for full, a 200 ms beep with a 1 s gap for empty, and off otherwise. The main loop only updates buzzer timing, so reception is never blocked.

7.3 Power Management and the V2.0 Event-Driven Algorithm

The outdoor node saves energy through the low-quiescent LDO, the MOSFET-switched sensor supply, a powered-down radio during sleep and the connectionless ESP-NOW link. Table 7.2 compares the running firmware with the event-driven V2.0 algorithm design.

Table 7.2 Implemented firmware compared with the V2.0 algorithm design

Aspect	Implemented firmware	V2.0 algorithm design
Wake interval	Fixed 30 s	LOW POWER mode every 5 min; ACTIVE mode every 1–2 min
Filtering	Single reading	Median of 3 readings after 50 ms sensor stabilisation
Transmission	Every cycle	Only if change ≥ delta threshold

Aspect	Implemented firmware	V2.0 algorithm design
Acknowledgement	Send-status callback	Indoor ACK within 100 ms, max 2 retries; 3 failures return to LOW POWER
Sleep mode	Timed light sleep	Deep sleep (design estimate ~20–70 µA; ~120 mA short active burst)
Level / buzzer	LED bands 25 / 50 / 80 %; buzzer for full and empty	Four equal 25 % bands; buzzer for full (and optionally a sudden drop)

8. PCB Design

8.1 Outdoor Node PCB (TWIN NODE V2.0)

V2.0 is a two-layer board of about 45 × 45 mm with the ESP-12F antenna overhanging the top edge, clear of copper. Compared with V1.0, mounting holes were increased from 3 to 4, the I²C expanders were removed, female headers were replaced by JST connectors, traces were widened and placement was optimised.

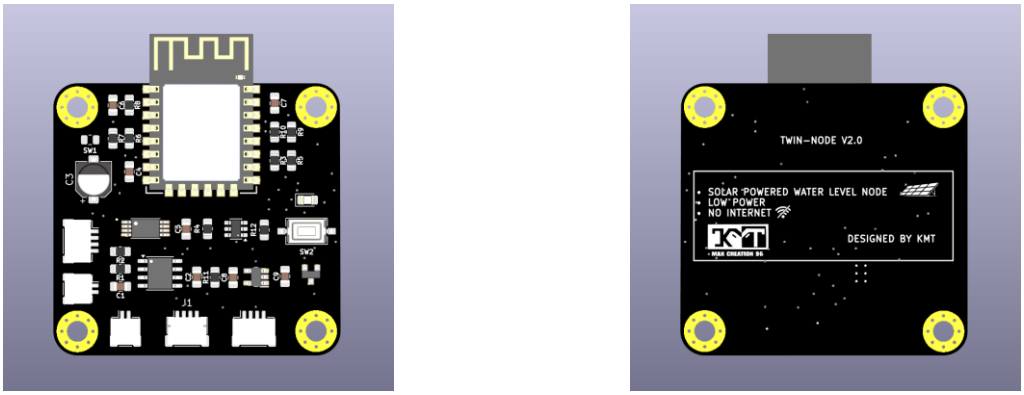


Figure 8.1 TWIN NODE V2.0 outdoor node – front and back (3D render)

8.2 Indoor Node PCB (TWIN NODE-INDOOR)

The indoor board is also two-layer with three mounting holes. The antenna overhangs the top edge, the micro-USB connector is at the bottom, the four LEDs, buzzer and battery pads are on the front, and the RST and BOOT buttons are at the sides. Gerber, drill, position, STEP and WRL files are in the repository.

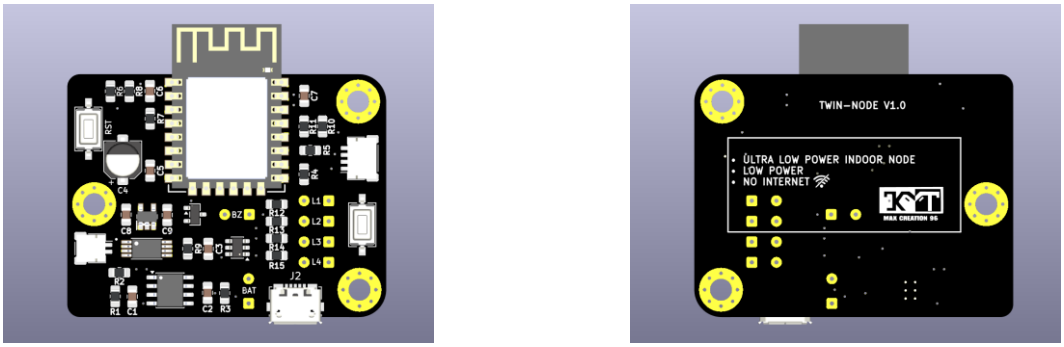


Figure 8.2 TWIN NODE-INDOOR – front and back (3D render)

9. Result and Discussion

9.1 Verification Results

Table 9.1 Design checks and hardware status

Board	DRC (violations / unconnected / footprint)	ERC	Status
Outdoor V2.0 (DRC 2026-01-04)	0 / 0 / 0	0 errors, 6 warnings	Fabricated; power-up, boot and reset validated
Indoor (DRC 2026-01-13)	0 / 0 / 0	0 errors, 10 warnings	Design and manufacturing files generated

The ERC warnings are not electrical faults; they are library-symbol mismatches (e.g. SI2301, LED) and pin-type warnings.

9.2 Fabrication Feedback

Testing of the V2.0 board showed that the external 10 kΩ pull-ups on GPIO0 (BOOT) and RST are unnecessary, since the existing 12 kΩ pull-ups suffice for normal boot and reliable reset. They can be removed in the next revision.

9.3 Functional Results

Table 9.2 Indoor response for different tank conditions (firmware behaviour)

Tank condition	LED	Buzzer
Full (≤ 5 cm; above 80 %)	FULL	Continuous at full flag, else off
51 – 80 %	HIGH	Off
26 – 50 %	MEDIUM	Off
1 – 25 %	LOW	Off
Empty (distance > 25 cm)	LOW	200 ms beep, 1 s gap, repeating
No echo / sensor error	All OFF	Off (no false empty alarm)

9.4 Discussion

- The direct ESP-NOW link needs no router or internet, so it works during outages with no running cost.
- MAC filtering and packet-size checking prevent other ESP-NOW devices from disturbing the indication.
- A no-echo reading never triggers the empty alarm, avoiding false alarms from splashes or foam.
- The 25 cm and 5 cm distances are firmware constants to be set for the real tank, and the sensor's blind zone must be respected when mounting.
- LED and buzzer thresholds differ slightly, and the indoor node keeps its radio listening continuously; both can be refined.

10. Conclusion and Future Work

10.1 Conclusion

TWIN NODE delivers a complete internet-free water level monitoring solution. The outdoor node (V2.0) senses the level with a power-gated ultrasonic sensor, runs from a protected solar-charged Li-ion cell and sleeps between readings; the indoor node gives a four-step LED display and audible alerts. Both boards were designed as two-layer PCBs with zero DRC violations, the outdoor board was fabricated and powered on successfully, and the firmware uses callbacks, non-blocking timing, MAC filtering and size validation.

10.2 Future Work – Motor Controller

The next stage is automatic pump control using the MOTOR CONTROLLER board already designed in the repository, an ESP-01 (ESP8266) based module that switches the pump through a relay using ESP-NOW.

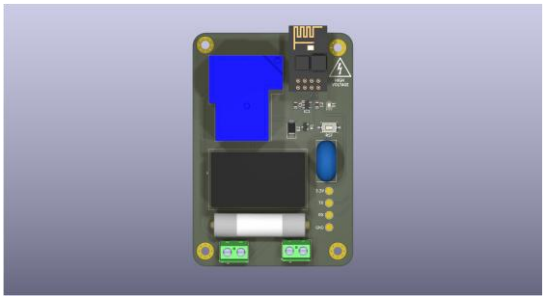


Figure 10.1 Motor controller PCB (3D render)

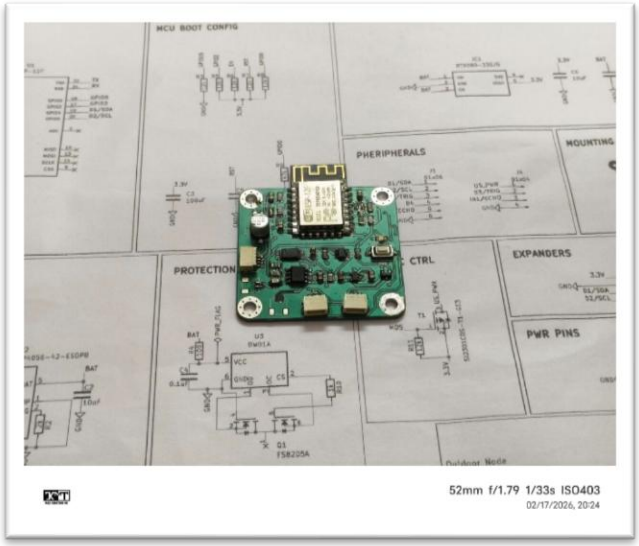
The board uses an HLK-PM01 230 V AC to 5 V module, an RT9080-3.3 LDO, and an SLA-5VDC-SL-C relay driven by an SI2300 MOSFET with a flyback diode. The mains side has a fuse and varistor, and the design checks are clean (DRC 0 violations; ERC 0 errors and 0 warnings).

Planned integration: pump ON/OFF commands are sent over ESP-NOW, switching on at low or empty level and off at full level, with a fail-safe that switches the pump off if level data stops arriving.

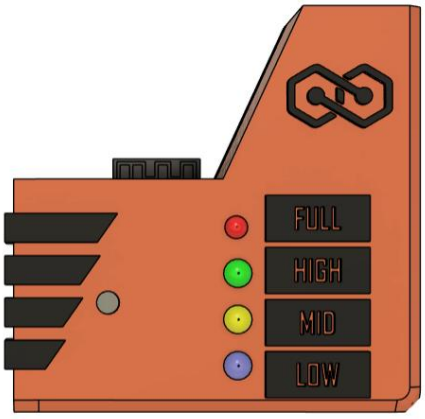
10.3 Other Improvements

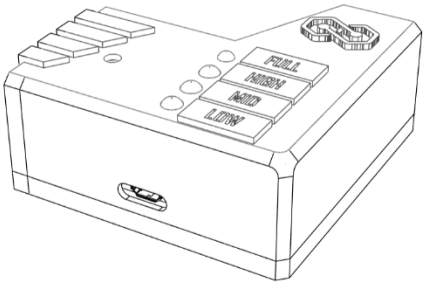
- Adopt the V2.0 event-driven algorithm: median filtering, change-based transmission, ACK with retries and deep sleep.
- Remove the redundant 10 kΩ pull-ups on GPIO0 and RST.
- Send battery voltage in the packet and add a low-power receive scheme to the indoor node.
- Add mobile-app integration, cloud logging, adjustable calibration, ESP-NOW encryption and a weatherproof enclosure.

11.Assembly



12.Enclosure





13. References

[1] Espressif Systems, “ESP8266EX Datasheet” and “ESP-NOW User Guide”.

[2] NanJing Top Power ASIC Corp., “TP4056 1A Standalone Linear Li-Ion Battery Charger Datasheet”.

[3] DW01A battery protection IC and FS8205A dual N-channel MOSFET datasheets.

[4] Richtek Technology, “RT9080 Low Quiescent Current LDO Regulator Datasheet”.

[5] Hi-Link, “HLK-PM01 AC-DC 5 V 3 W Power Module Specification”.

[6] Arduino core for ESP8266 documentation (ESP8266WiFi, ESP-NOW and light-sleep examples).

[7] KiCad EDA documentation, https://www.kicad.org.

[8] Maathes Thilak K, “TWIN-NODE” project repository, https://github.com/MAATHES-THILAK-K/TWIN-NODE.